Supplementary material

Title

Evaluating Agreement of Large Language Models with Clinician Based Cause-of Death Determination in Malawi

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Supplementary Section A: Models and Prompts

Supplementary Table 1 Open AI models used in the CoD pipeline.

Task / model used	Model name, date and standard parameters	Parameters used
Reconstructed narrative from questionnaire	**GPT-4.1-mini** Date: April 14, 2025 Temperature: 1 Top p: 1 Standard pricing: input tokens $0.40 per 1 million, $1.60 per one million Batch processing: 50% discount for non-urgent tasks Prompt cache input read: $0.10 per million tokens.	Standard
Causes of Death in English	**GPT-5.2** Date: December 11, 2025 Temperature: 1 Top p: 1 Standard pricing: input tokens $1.75 per 1 million, $14.00 per one million Batch processing: 50% discount for non-urgent tasks Prompt cache input read: $0.175 per million tokens.	Standard
Mapping to MEIRU codes	**GPT-4.1** Date: April 14, 2025 Temperature: 1 Top p: 1 Standard pricing: input tokens $2.00 per 1 million, $8.00 per one million Batch processing: 50% discount for non-urgent tasks Prompt cache input read: $0.50 per million tokens.	Standard

Supplementary Table 2 Prompts used.

Prompt purpose	Prompt
To generate a narrative from questionnaire-as-text.	# Instructions Rewrite the following verbal autopsy clinical narrative into a clearer, shorter paragraph. # Rules Do not add any new information. Do not infer a cause of death. Do not use outside knowledge. Remove repetition. Keep all clinically relevant facts in the same order as the original text. Keep uncertain, reported, or respondent-stated information as reported. Improve grammar and flow. Round illness duration to the nearest whole day where appropriate. Output only the rewritten paragraph.

To generate a cause of death determination in English based on the input.	# Role and Objective You are assisting with verbal autopsy cause-of-death coding using the MEIRU cause-of-death coding system. # Instructions Infer the causes of death based on the following narrative of a relative: 1. First identify the most immediate/direct cause of death from the narrative. If one cannot be established with certainty, leave blank. 2. Then identify the underlying cause of death. This field is mandatory and must always be completed. 3. Finally identify any contributory causes of death that may have contributed but were not the main underlying cause. # Rules - Give the underlying, direct and contributory causes. - Causes of death separated by semicolon. - Causes of death are in English. - The explanation must be in English, formatted as JSON, with each item structured using factor, cause_of_death, and impact extent, and based only on information present in the narrative.

	# Role and Objective
	Your task is to map the causes of death given to the single best MEIRU code. You will receive underlying, direct and contributory causes of death.
	
	# Rules:
	- Use only the MEIRU codes provided in the codebook below.
	- Do not invent new codes.
	- Do not use outside information.
	- Base the decision only on the causes of death given.
	- Return only valid JSON.
	
	Return JSON in this format:
	{
	"direct_cause_of_death": {
	"selected_full_code": "",
	"selected_description": "",
	"confidence": "high | medium | low",
	"supporting_evidence": [],
	"brief_explanation": ""
	},
	"underlying_cause_of_death": {
	"selected_full_code": "",
	"selected_description": "",
	"confidence": "high | medium | low",
	"supporting_evidence": [],
	"brief_explanation": ""
	},
	"contributory_causes_of_death": [

	{
	"selected_full_code": "",
	"selected_description": "",
	"confidence": "high | medium | low",
	"supporting_evidence": [],
	"brief_explanation": ""
	}
	],
	"overall_uncertainty": "",
	"coding_notes": ""
	}`;

MERU Codebook provided for the prompts	00 - 01 - 00 | Unspecifiable: No information (VAQ not available) 00 - 02 - 00 | Unspecifiable: Specific information missing on VAQ 00 - 03 - 00 | Unspecifiable: No significant pathology that would explain the death 00 - 04 - 00 | Unspecifiable: Multiple significant pathology, single CoD can't be isolated 01 - 00 - 00 | Communicable disease: Unspecifiable 01 - 01 - 00 | Communicable disease: Acute febrile illness: Unspecifiable 01 - 01 - 01 | Communicable disease: Acute febrile illness: Malaria 01 - 01 - 02 | Communicable disease: Acute febrile illness: Menigitis 01 - 01 - 03 | Communicable disease: Acute febrile illness: Measles 01 - 01 - 04 | Communicable disease: Acute febrile illness: Pneumonia 01 - 01 - 05 | Communicable disease: Acute febrile illness: Diarrhoea with fever 01 - 01 - 06 | Communicable disease: Acute febrile illness: Sepsis 01 - 01 - 99 | Communicable disease: Acute febrile illness: Other specific unlisted 01 - 03 - 00 | Communicable disease: Hepatitis 01 - 04 - 00 | Communicable disease: TB/AIDS: Unspecifiable 01 - 04 - 01 | Communicable disease: TB/AIDS: Pulmonary TB 01 - 04 - 02 | Communicable disease: TB/AIDS: AIDS 01 - 04 - 03 | Communicable disease: TB/AIDS: Extrapulmonary TB 01 - 04 - 99 | Communicable disease: TB/AIDS: Other specific unlisted 01 - 05 - 00 | Communicable disease: Diarrhoeal disease without fever 01 - 06 - 00 | Communicable disease: Tetanus (exc. neonatal tetanus) 01 - 07 - 00 | Communicable disease: Rabies 01 - 99 - 00 | Communicable disease: Other specific unlisted 02 - 00 - 00 | Direct maternal cause: Unspecifiable 02 - 01 - 00 | Direct maternal cause: Abortion 02 - 02 - 00 | Direct maternal cause: Eclampsia 02 - 03 - 00 | Direct maternal cause: Ante/postpartum haemorrhage 02 - 04 - 00 | Direct maternal cause: Obstructed labour 02 - 05 - 00 | Direct maternal cause: Puerperal sepsis 02 - 06 - 00 | Direct maternal cause: Anaemia in pregnancy 02 - 99 - 00 | Direct maternal cause: Other specific unlisted 03 - 00 - 00 | Non communicable disease: Unspecifiable 03 - 01 - 00 | Non communicable disease: Cardiovascular disorder: Unspecifiable 03 - 01 - 01 | Non communicable disease: Cardiovascular disorder: Hypertension

	03 - 01 - 02 | Non communicable disease: Cardiovascular disorder: Congestive heart disease 03 - 01 - 03 | Non communicable disease: Cardiovascular disorder: Ischaemic heart disease 03 - 01 - 04 | Non communicable disease: Cardiovascular disorder: Cerebro vascular disease 03 - 01 - 99 | Non communicable disease: Cardiovascular disorder: Other specific unlisted 03 - 02 - 00 | Non communicable disease: Respiratory disorder: Unspecifiable 03 - 02 - 01 | Non communicable disease: Respiratory disorder: Chronic obstructive pulmonary disease 03 - 02 - 02 | Non communicable disease: Respiratory disorder: Asthma 03 - 02 - 99 | Non communicable disease: Respiratory disorder: Other specific unlisted 03 - 03 - 00 | Non communicable disease: Gastro intestinal disorder: Unspecifiable 03 - 03 - 01 | Non communicable disease: Gastro intestinal disorder: Peptic ulcer disease 03 - 03 - 02 | Non communicable disease: Gastro intestinal disorder: Liver cirrhosis 03 - 03 - 03 | Non communicable disease: Gastro intestinal disorder: Acute abdomen including obstruction 03 - 03 - 99 | Non communicable disease: Gastro intestinal disorder: Other specific unlisted 03 - 04 - 00 | Non communicable disease: Central nervous system disorder: Unspecifiable 03 - 04 - 01 | Non communicable disease: Central nervous system disorder: Mental/behavioural disorder 03 - 04 - 02 | Non communicable disease: Central nervous system disorder: Epilepsy 03 - 04 - 99 | Non communicable disease: Central nervous system disorder: Other specific unlisted 03 - 05 - 00 | Non communicable disease: Endocrine disorders: Unspecifiable 03 - 05 - 01 | Non communicable disease: Endocrine disorders: Diabetes 03 - 05 - 99 | Non communicable disease: Endocrine disorders: Other specific unlisted 03 - 06 - 00 | Non communicable disease: Neoplasm: Unspecifiable 03 - 06 - 01 | Non communicable disease: Neoplasm: Breast 03 - 06 - 02 | Non communicable disease: Neoplasm: Cervix/uterus 03 - 06 - 03 | Non communicable disease: Neoplasm: Liver 03 - 06 - 04 | Non communicable disease: Neoplasm: Gastro intestinal tract including abdominal but excluding liver 03 - 06 - 05 | Non communicable disease: Neoplasm: Lung 03 - 06 - 06 | Non communicable disease: Neoplasm: Oral 03 - 06 - 07 | Non communicable disease: Neoplasm: Oesophagus 03 - 06 - 99 | Non communicable disease: Neoplasm: Other specific unlisted 03 - 07 - 00 | Non communicable disease: Genito urinary disorders: Unspecifiable

	03 - 07 - 01 | Non communicable disease: Genito urinary disorders: Kidney disorder 03 - 07 - 99 | Non communicable disease: Genito urinary disorders: Other specific unlisted 03 - 08 - 00 | Non communicable disease: Anaemia: Unspecifiable 03 - 08 - 01 | Non communicable disease: Anaemia: Caused by chronic communicable disease 03 - 08 - 99 | Non communicable disease: Anaemia: Other specific unlisted 03 - 09 - 00 | Non communicable disease: Nutritional disorder: Unspecifiable 03 - 09 - 01 | Non communicable disease: Nutritional disorder: Malnutrition 03 - 09 - 99 | Non communicable disease: Nutritional disorder: Other specific unlisted 03 - 99 - 00 | Non communicable disease: Other specific unlisted 04 - 00 - 00 | External cause: Unspecifiable 04 - 01 - 00 | External cause: Transport 04 - 02 - 00 | External cause: Fall 04 - 03 - 00 | External cause: Drowning/submersion 04 - 04 - 00 | External cause: Exposure to smoke/fire/flames 04 - 05 - 00 | External cause: Poisoning/exposure to noxious substance 04 - 06 - 00 | External cause: Use of weapon 04 - 07 - 00 | External cause: Contact with venomous animals and plants 04 - 08 - 00 | External cause: Hanging 04 - 09 - 00 | External cause: Choking 04 - 99 - 00 | External cause: Other specific unlisted 05 - 00 - 00 | Causes specific to infancy: Unspecifiable 05 - 01 - 00 | Causes specific to infancy: Miscarriage/abortion 05 - 02 - 00 | Causes specific to infancy: Stillbirth 05 - 03 - 00 | Causes specific to infancy: Prematurity/low birth weight 05 - 04 - 00 | Causes specific to infancy: Congenital abnormality 05 - 05 - 00 | Causes specific to infancy: Birth injury/asphyxia 05 - 06 - 00 | Causes specific to infancy: Neonatal sepsis 05 - 08 - 00 | Causes specific to infancy: Sudden infant death syndrome 05 - 09 - 00 | Causes specific to infancy: Neonatal tetanus 05 - 99 - 00 | Causes specific to infancy: Other specific unlisted 99 - 00 - 00 | Other specific unlisted

Supplementary Section B: Examples of inputs and outputs

Supplementary Table 3 An example of original narrative, questionnaire-as-text and extended questionnaire-as-text portion listing variables without data.

Type of input	Example
Original narrative	Anonymised narrative is: The main problems were diarrhoea and vomiting. This took one week. Parents tried way to heal the child but not improvement. The child died.
Questionnaire-as-text	The deceased was a 0-year-old male. The deceased had know cod. Respondent COD: DIARRHOEA AND VOMITING. TB was no. Duration illness days: 7.0. Fever was no. Cough was no. Diarrhoea was no. The deceased had vomit. Vomit days: 7.0. Stiff neck was no. Unconscious >24 hrs was no. Convulsions was no. Dry scaly skin was no. Skin rash was no. Any lumps was no. The deceased had weight loss. Severely thin was no. Yellow eyes was no. Injury was no. The deceased had attend hosp. Adult Child or Neonate was vac (child).
Extended part of the questionnaire-as-text listing blank variables	The following information was not recorded or is unknown: Hospital told COD, What hosp said was COD, Do you wish to carry out the full or short VA?, Where TB treated, TB treatment yr, HIV/AIDS, Pos malaria, Neg malaria, Measles, Hypertension, Heart disease, Diabetes, Asthma, Epilepsy, Cancer, COPD, Dementia, Depression, Stroke, Sickle cell, Kidney disease, Liver disease, Any diagnosis of covid-19 in last month?, Was ever tested for covid-19?, Is the date of covid-19 test known?, Date of covid, What was the result of the covid-19 test?, season of death, Die suddenly, Days had fever, Fever severe, Fever continuously/on & off, Fever only at night, Chills/rigor, Night sweats, Did have a runny nose?, Was sneezing?, Did have a blocked nose?, Did have a sore throat?, Did have a hoarse voice?, Days had cough, Cough productive, Sputum exam, Sputum result, Cough blood, Whooping, Fast breathing days, Breathlessness, Breathlessness days, No daily routine breathless, Breathless lying flat, Chest pain, Chest pain duration, Did have tightness in the chest?, Did have decrease in appetite?, Did have loss of sense of smell?, Did have loss of sense of taste?, Diarrhoea days, Diarrhoea times day, What stool look like, Vomit blood, Abdominal pain, Days had abdominal pain, Where Abdominal pain, Abdomen distension, Abdominal distension days, Abdominal mass, Abdominal mass days, Constipation, Headache, Headache days, Stiff neck days, Confusion, Confusion days, Unconscious sudden, Confusion before unconscious, Convulsions days, Convulsions minutes, Unconscious after convulsions, Did experience dizziness?, Did have difficulty sleeping?, Did feel more tired than normal?, Did have severe fatigue (e.g. inability to get out of bed)?, Difficulty urinating, No urine, Urinate more often, Bloody urine, Abscess, Abscess feet, Itching of skin, Rash days, What rash look like, Rash all over, Skin spit/peel, Ever shingles, Gen. body swelling, Face swelling, Feet swelling, Feet swelling days, Altered sensation hands/feet, Did have numbness or tingling somewhere in the body?, Did have a feeling of heaviness in arms or legs?, Did have achy muscles?, Lumps days, Lump mouth, Lump neck, Lump armpit, Lump groin, Lump other, Bleed from orifice, Body stiffness, One side paralysis, One side paralysis days, Swallow pain, Swallow pain days, Did have sore eyes?, Changing hair colour, Anaemic, Drink more water, Injury days, Injury accidental or intentional, Alcohol contribute

	to injury, Type of injury, Type of RTA, Counterpart in RTA, Animal/insect bite type, Assault type, Bleeding between periods, Menopause, Bleeding after menopause, Lump breast, Vaginal tumours, Pregnant, Gave birth <6wks before, Preg lasted <6m, Gave birth 6wk-1yr before, Drink alcohol, Smoke, Treated, ORT, IV -, Blood transfusion, NG tube, Antibiotic injection, Operation, Operation >=1m, Discharged v ill, >2hr to get to hospital, Doubts med care needed, Trad med, Phone used, Care too expensive, Motor vehicle to hosp, Admission probs, Treatment probs, Probs getting meds, HIV test result, Mother ever HIV test, Mother HIV test result, HIV tested where, HIV test date estimated, HIV test date (dec or mother), Ever taken anti-retrovirals, ART start date estimated, ART start date, Health doc reviewed.

Supplementary Table 4 Example of LLM CoD outputs.

Output type	Example
CoD determination in English.	Direct cause of death: Dehydration Underlying cause of death: Acute gastroenteritis (diarrhoeal disease) Contributory causes of death: Vomiting; Weight loss/malnutrition

CoD using MEIRU codes	{
	"direct_cause_of_death": {
	"selected_full_code": "01 - 05 - 00",
	"selected_description": "Communicable disease: Diarrhoeal disease without fever",
	"confidence": "high",
	"supporting_evidence": [
	"Direct cause of death stated as dehydration",
	"Underlying cause is acute gastroenteritis (diarrhoeal disease)",
	"No mention of fever in causes"
	],
	"brief_explanation": "Dehydration commonly results from diarrhoeal illness. As there is no explicit mention of fever, the code for diarrhoeal disease without fever is the best fit."
	},
	"underlying_cause_of_death": {
	"selected_full_code": "01 - 05 - 00",
	"selected_description": "Communicable disease: Diarrhoeal disease without fever",
	"confidence": "high",
	"supporting_evidence": [
	"Underlying cause clearly stated as acute gastroenteritis (diarrhoeal disease)",
	"No mention of fever, so 'without fever' is appropriate"
	],
	"brief_explanation": "Acute gastroenteritis is a diarrhoeal disease, and with no mention of fever, this code matches directly."
	},
	"contributory_causes_of_death": [
	{
	"selected_full_code": "03 - 09 - 01",
	"selected_description": "Non communicable disease: Nutritional disorder:

	Malnutrition", "confidence": "high", "supporting_evidence": ["Weight loss/malnutrition specifically mentioned as a contributory cause"], "brief_explanation": "Malnutrition is listed in the codebook and matches the contributory cause stated." }], "overall_uncertainty": "low", "coding_notes": "No mention of fever, so diarrhoeal disease without fever selected. Vomiting, a sign, is not directly mapped; weight loss/malnutrition mapped to malnutrition code. High confidence due to clear matching terms." }

Supplementary Section C: Statistics for the MEIRU Data

Supplementary Table 5 Characteristics of the MEIRU VA dataset. Percentages for record completeness, VA type and sex are calculated using the respective column totals. Percentages for age groups are calculated within the corresponding VA type. Adult verbal autopsies are defined as those aged 12 years and above according to the MEIRU classification. Neonatal deaths and stillbirths are grouped together under the MEIRU classification. Percentages may not sum to 100% because of rounding.

Characteristic		**All records, n = 5150**	**Summarised records, n = 4122**
VA type	Adult	3,487 (68%)	2,684 (65%)
	Child	1,049 (20%)	934 (23%)
	Neonate/stillbirth	614 (12%)	504 (12%)
Sex entire dataset	Female	2,697 (52%)	2,129 (52%)
	Male	2,453 (48%)	1,993 (48%)
Sex within Adult	Female	1,794 (51%)	1,347 (50%)
	Male	1,693 (49%)	1,337 (50%)
Sex within Child	Female	561 (54%)	503 (54%)
	Male	488 (47%)	431 (46%)
Sex within Neonate/stillbirth	Female	342 (58%)	279 (55%)
	Male	272 (44%)	225 (45%)
Age group within Adult	12-14	40 (1%)	24 (1%)
	15-24	236 (7%)	179 (7%)
	25-44	931 (27%)	815 (30%)
	45-64	786 (23%)	612 (23%)
	65+	1,494 (43%)	1,054 (39%)
Age group within Child	0 -1	481 (46%)	432 (46%)
	2-4	356 (34%)	323 (35%)
	5-8	147 (14%)	127 (14%)
	9-11	50 (5%)	39 (4%)
	Out of range	15 (1%)	13 (1%)

Supplementary Table 6 Distribution of the most frequently assigned Level 1, Level 2 and Level 3 causes of death, by type. Level 1 calculations include unspecifiable clinician causes, while Level 2 exclude these. The 15 most frequent categories were selected independently at Levels 2 and 3 according to the proportion of specified underlying causes of death. The corresponding proportions for direct and contributory causes are shown for the same categories. At Level 3, codes ending in 00 indicate that the broader cause category was assigned, but specific diagnoses were not recorded.

Panel A. Level 1 causes

Level 1 cause of death category	**Underlying, n =** 4,122	**Direct, n =** 333	**Contributory, n =** 260
00 -Unspecifiable	256 (6%)	3 (1%)	3 (1%)
01-Communicable disease	1781 (43%)	156 (47%)	42 (16%)
02-Direct maternal cause	55 (1%)	4 (1%)	1 (0%)
03-Non communicable disease	1243 (30%)	138 (41%)	202 (78%)
04-External cause	278 (7%)	14 (4%)	4 (2%)
05-Causes specific to infancy	509 (12%)	18 (5%)	8 (3%)
Non-blanks	4122 (100%)	333 (100%)	260 (100%)

Level 2 causes

Code	**Cause**	**Underlying**	**Direct**	**Contributory**
01-04	TB/AIDS	23.5%	11.8%	12.1%
01-01	Acute febrile illness	19.9%	29.7%	3.1%
03-01	Cardiovascular disorder	11.7%	25.5%	40.9%
03-03	Gastrointestinal disorder	5.4%	6.4%	0.4%
05-05	Birth injury/asphyxia	5.3%	0.3%	1.2%
03-06	Neoplasm	5.0%	3.3%	0.8%
05-03	Prematurity/low birth weight	3.8%	0.3%	1.6%
05-06	Neonatal sepsis	2.6%	4.8%	0.0%
03-04	Central nervous system disorder	2.0%	1.2%	5.1%
03-05	Endocrine disorders	1.9%	0.6%	14.8%
04-01	Transport	1.8%	0.6%	0.4%
03-09	Nutritional disorder	1.6%	0.0%	3.9%
04-03	Drowning/submersion	1.4%	0.9%	0.0%
04-05	Poisoning/exposure to noxious substance	1.4%	1.2%	0.4%
01-05	Diarrhoeal disease without fever	1.3%	0.6%	0.4%

Panel B. Level 3 causes

Code	**Cause**	**Underlying**	**Direct**	**Contributory**
01-04-02	AIDS	20.7%	0.6%	10.5%
01-01-00	Acute febrile illness — unspecified	5.5%	3.9%	0.8%
01-01-04	Pneumonia	5.4%	7.6%	0.4%
01-01-01	Malaria	5.3%	3.6%	1.9%
05-05-00	Birth injury/asphyxia	5.3%	0.3%	1.2%
03-01-02	Congestive heart disease	3.8%	10.0%	0.4%
05-03-00	Prematurity/low birth weight	3.8%	0.3%	1.6%
03-01-04	Cerebrovascular disease	3.6%	12.7%	8.6%
03-01-01	Hypertension	3.2%	0.3%	31.9%

03-03-00	Gastrointestinal disorder — unspecified	2.6%	4.2%	0.4%
05-06-00	Neonatal sepsis	2.6%	4.8%	0.0%
03-05-01	Diabetes	1.9%	0.3%	14.8%
01-01-05	Diarrhoea with fever	1.8%	2.7%	0.0%
04-01-00	Transport	1.8%	0.6%	0.4%
03-06-02	Cervix/uterus neoplasm	1.6%	0.3%	0.4%

Supplementary Table 7 Clinician causes by age groups for Top 10 underlying causes

Panel A. Adult clinician determinations, N=2,453.

Cause	Total N	12–14	15–24	25–44	45–64	65+
TB/AIDS	767	14%	23%	54%	44%	6%
Cardiovascular disorder	449	5%	4%	1%	12%	39%
Acute febrile illness	254	14%	13%	8%	7%	14%
Gastrointestinal disorder	196	5%	7%	9%	7%	8%
Neoplasm	183	9%	4%	3%	9%	11%
Endocrine disorders	73	5%	1%	2%	3%	5%
Central nervous system disorder	65	14%	8%	3%	2%	2%
Transport	60	0%	8%	4%	2%	1%
Poisoning/exposure to noxious substance	48	0%	4%	3%	3%	1%

Denominators: 12–14, N=22; 15–24, N=170; 25–44, N=769; 45–64, N=582; and 65+, N=910.

Panel B. Child clinician determination, N = 912

Cause	Total N	0–1	2–4	5–8	9–11	Out of range
Acute febrile illness	501	58%	60%	43%	36%	17%
TB/AIDS	142	18%	11%	17%	15%	33%
Nutritional disorder	46	4%	7%	6%	3%	0%
Diarrhoeal disease without fever	36	6%	3%	0%	0%	0%
Drowning/submersion	25	0%	3%	8%	5%	8%
Anaemia	23	2%	3%	2%	5%	0%
Congenital abnormality	17	3%	1%	2%	0%	0%
External cause, unspecified	14	1%	2%	1%	3%	8%
Gastrointestinal disorder	12	0%	1%	3%	5%	8%
Central nervous system disorder	12	0%	2%	0%	8%	0%

Denominators: 0–1, N=420; 2–4, N=315; 5–8, N=126; 9–11, N=39; and out of range, N=12.

Panel C. Neonate/stillbirth clinician determinations, N=501

Cause	Total N	Neonates/Stillborths
Birth injury/asphyxia	205	41%
Prematurity/low birth weight	145	29%
Neonatal sepsis	96	19%
Neonatal cause, unspecified	20	4%
Acute febrile illness	16	3%
Congenital abnormality	11	2%
Sudden infant death syndrome	2	0%
Nutritional disorder	2	0%
External cause, unspecified	1	0%
Antepartum haemorrhage	1	

Supplementary Section D: Characteristics of the GPT model determinations of Cause of Death

Supplementary Table 8 Flexible agreement for underlying and direct causes. Eligible clinician records at Level 1,2,3 are 4,122, 3,869, 3,869 for pooled causes.

Experiment	Level	Any-code overlap, %	Mean Jaccard (95% CI)	Exact-set agreement, %
EXP1	1	88.0	0.795 (0.784–0.805)	71.0
EXP1	2	78.7	0.640 (0.627–0.653)	49.8
EXP1	3	63.2	0.480 (0.466–0.492)	33.7
EXP2	1	82.3	0.727 (0.715–0.739)	63.2
EXP2	2	64.2	0.532 (0.518–0.546)	42.7
EXP2	3	49.7	0.392 (0.378–0.405)	29.4
EXP3	1	79.8	0.699 (0.686–0.711)	60.1
EXP3	2	69.0	0.551 (0.537–0.564)	41.6
EXP3	3	55.0	0.410 (0.397–0.423)	27.9
EXP4	1	82.0	0.718 (0.706–0.730)	61.8
EXP4	2	64.5	0.521 (0.507–0.535)	40.1
EXP4	3	49.9	0.379 (0.366–0.392)	26.7
EXP5	1	87.8	0.787 (0.776–0.798)	69.7
EXP5	2	79.0	0.625 (0.613–0.637)	46.4
EXP5	3	64.7	0.471 (0.458–0.483)	30.3

Supplementary Table 9 Breakdown of clinician determinations for the summarised dataset of underlying causes of death by VA type for EXP1. Denominators: N = 4122, Adult = 2,684, Child n=934, neonate/stillbirth n = 504. Showing full codes at Level 1 and top 15 at Level 2 and Level 3. Percentages are column percentages rounded to whole numbers. We use this frequency analysis to compare LLM determinations against clinician CoDs, Level 1 determinations remain included as 00, 00-00, and 00-00-00.

Level / code		CoD name	Adult	Child	Neonates/Stillborths	Total
1	01	Communicable disease	1071 (44%)	694 (76%)	16 (3%)	1781 (43.2%)
	03	Non communicable disease	1119 (46%)	121 (13%)	3 (1%)	1243 (30.2%)
	05	Causes specific to infancy	0 (0%)	30 (3%)	479 (96%)	509 (12.3%)
	04	External cause	210 (9%)	67 (7%)	1 (0%)	278 (6.7%)
	00	Unspecifiable	231 (9%)	22 (2%)	3 (1%)	256 (6.2%)
	02	Direct maternal cause	53 (2%)	0 (0%)	2 (0%)	55 (1.3%)
2	01-04	TB/AIDS	767 (29%)	142 (15%)	0 (0%)	909 (22%)

	01-01	Acute febrile illness	254 (9%)	501 (54%)	16 (3%)	771 (19%)
	03-01	Cardiovascular disorder	449 (17%)	4 (0%)	0 (0%)	453 (11%)
	00-00	Unspecifiable	231 (9%)	22 (2%)	3 (1%)	256 (6%)
	03-03	Gastrointestinal disorder	196 (7%)	12 (1%)	1 (0%)	209 (5%)
	05-05	Birth injury/asphyxia	0 (0%)	1 (0%)	205 (41%)	206 (5%)
	03-06	Neoplasm	183 (7%)	10 (1%)	0 (0%)	193 (5%)
	05-03	Prematurity/low birth weight	0 (0%)	0 (0%)	145 (29%)	145 (4%)
	05-06	Neonatal sepsis	0 (0%)	5 (1%)	96 (19%)	101 (2%)
	03-04	Central nervous system disorder	65 (2%)	12 (1%)	0 (0%)	77 (2%)
	03-05	Endocrine disorders	73 (3%)	0 (0%)	0 (0%)	73 (2%)
	04-01	Transport	60 (2%)	9 (1%)	0 (0%)	69 (2%)
	03-09	Nutritional disorder	12 (0%)	46 (5%)	2 (0%)	60 (1%)
	04-03	Drowning/submersion	30 (1%)	25 (3%)	0 (0%)	55 (1%)
	04-05	Poisoning/exposure to noxious substance	48 (2%)	6 (1%)	0 (0%)	54 (1%)
3	01-04-02	AIDS	675 (25%)	126 (13%)	0 (0%)	801 (19%)
	00-00-00	Unspecifiable, Level 1 only	231 (9%)	22 (2%)	3 (1%)	256 (6%)
	01-01-00	Acute febrile illness, unspecified	76 (3%)	136 (15%)	2 (0%)	214 (5%)
	01-01-04	Pneumonia	75 (3%)	122 (13%)	12 (2%)	209 (5%)
	01-01-01	Malaria	39 (1%)	166 (18%)	1 (0%)	206 (5%)
	05-05-00	Birth injury/asphyxia	0 (0%)	1 (0%)	205 (41%)	206 (5%)
	03-01-02	Congestive heart disease	145 (5%)	0 (0%)	0 (0%)	145 (4%)
	05-03-00	Prematurity/low birth weight	0 (0%)	0 (0%)	145 (29%)	145 (4%)
	03-01-04	Cerebrovascular disease	139 (5%)	0 (0%)	0 (0%)	139 (3%)
	03-01-01	Hypertension	124 (5%)	0 (0%)	0 (0%)	124 (3%)
	03-03-00	Gastrointestinal disorder, unspecified	95 (4%)	5 (1%)	1 (0%)	101 (2%)
	05-06-00	Neonatal sepsis	0 (0%)	5 (1%)	96 (19%)	101 (2%)
	03-05-01	Diabetes	72 (3%)	0 (0%)	0 (0%)	72 (2%)
	01-01-05	Diarrhoea with fever	15 (1%)	55 (6%)	0 (0%)	70 (2%)

04-01-00	Transport	60 (2%)	9 (1%)	0 (0%)	69 (2%)

Supplementary Table 10 Breakdown of determinations of underlying causes of death in EXP1 for all summarised records by VA type. We show full codes breakdown at Level 1 and top 15 for Level 2 and Level 3 codes. The denominators are N=4122, Adult = 2,684, Child = 934 and Neonates = 504. Percentages are column percentages.

Level / Code		CoD name	Adult	Child	Neonate/ stillbirth	Total
1	01	Communicable disease	1,129 (42%)	688 (74%)	30 (6%)	1,847 (45%)
	03	Non communicable disease	1,221 (45%)	150 (16%)	3 (1%)	1,374 (33%)
	05	Causes specific to infancy	1 (0%)	30 (3%)	403 (80%)	434 (11%)
	04	External cause	204 (8%)	63 (7%)	6 (1%)	273 (7%)
	02	Direct maternal cause	54 (2%)	0 (0%)	56 (11%)	110 (3%)
	00	Unspecifiable	75 (3%)	3 (0%)	6 (1%)	84 (2%)
2						
	01-01	Acute febrile illness	371 (14%)	519 (56%)	30 (6%)	920 (22%)
	01-04	TB/AIDS	681 (25%)	82 (9%)	0 (0%)	763 (19%)
	03-01	Cardiovascular disorder	513 (19%)	11 (1%)	0 (0%)	524 (13%)
	03-06	Neoplasm	226 (8%)	12 (1%)	0 (0%)	238 (6%)
	03-03	Gastrointestinal disorder	209 (8%)	9 (1%)	1 (0%)	219 (5%)
	05-03	Prematurity/low birth weight	0 (0%)	5 (1%)	141 (28%)	146 (4%)
	05-05	Birth injury/asphyxia	0 (0%)	1 (0%)	131 (26%)	132 (3%)
	01-05	Diarrhoeal disease without fever	39 (1%)	79 (8%)	0 (0%)	118 (3%)
	05-06	Neonatal sepsis	0 (0%)	5 (1%)	95 (19%)	100 (2%)
	03-04	Central nervous system disorder	64 (2%)	15 (2%)	0 (0%)	79 (2%)
	03-09	Nutritional disorder	13 (0%)	63 (7%)	1 (0%)	77 (2%)
	04-01	Transport	63 (2%)	9 (1%)	0 (0%)	72 (2%)
	02-04	Obstructed labour	15 (1%)	0 (0%)	55 (11%)	70 (2%)
	03-02	Respiratory disorder	57 (2%)	10 (1%)	0 (0%)	67 (2%)
	03-05	Endocrine disorders	61 (2%)	0 (0%)	0 (0%)	61 (1%)
	02-02	Direct maternal cause: Eclampsia	4 (0%)	0 (0%)	1 (0%)	5 (0%)
	01-07	Communicable disease: Rabies	1 (0%)	2 (0%)	0 (0%)	3 (0%)
	04-04	External cause: Exposure to smoke/fire/flames	1 (0%)	2 (0%)	0 (0%)	3 (0%)
	02-06	Direct maternal cause: Anaemia in pregnancy	2 (0%)	0 (0%)	0 (0%)	2 (0%)

	00-04	Unspecifiable: Multiple significant pathology, single COD can't be isolated	1 (0%)	0 (0%)	0 (0%)	1 (0%)
	04-00	External cause: Unspecifiable	1 (0%)	0 (0%)	0 (0%)	1 (0%)
	05-09	Causes specific to infancy: Neonatal tetanus	0 (0%)	0 (0%)	1 (0%)	1 (0%)
3	01-04-02	AIDS	548 (20%)	63 (7%)	0 (0%)	611 (15%)
	01-01-01	Malaria	128 (5%)	264 (28%)	8 (2%)	400 (10%)
	01-01-04	Pneumonia	79 (3%)	138 (15%)	20 (4%)	237 (6%)
	03-01-01	Hypertension	199 (7%)	0 (0%)	0 (0%)	199 (5%)
	03-01-04	Cerebrovascular disease	153 (6%)	2 (0%)	0 (0%)	155 (4%)
	05-03-00	Prematurity/low birth weight	0 (0%)	5 (1%)	141 (28%)	146 (4%)
	05-05-00	Birth injury/asphyxia	0 (0%)	1 (0%)	131 (26%)	132 (3%)
	01-04-01	Pulmonary TB	113 (4%)	18 (2%)	0 (0%)	131 (3%)
	01-05-00	Diarrhoeal disease without fever	39 (1%)	79 (8%)	0 (0%)	118 (3%)
	03-03-02	Liver cirrhosis	102 (4%)	2 (0%)	0 (0%)	104 (3%)
	05-06-00	Neonatal sepsis	0 (0%)	5 (1%)	95 (19%)	100 (2%)
	03-01-02	Congestive heart disease	81 (3%)	1 (0%)	0 (0%)	82 (2%)
	01-01-02	Meningitis	52 (2%)	25 (3%)	0 (0%)	77 (2%)
	03-09-01	Malnutrition	13 (0%)	63 (7%)	1 (0%)	77 (2%)
	01-01-00	Acute febrile illness, unspecified	47 (2%)	26 (3%)	0 (0%)	73 (2%)

Section E: Flexible Agreement

Supplementary Table 11 Flexible agreement at Level 1 category Level using pooled clinician and experiment sets. All are at (95% CI)

Panel A. EXP1

Level 1 category	N	Any-code overlap, %	Mean Jaccard	Exact-set agreement, %
Communicable disease	1,781	97.8 (97.1–98.4)	0.63 (0.62–0.64)	30.0 (28.0–32.1)
Direct maternal cause	55	94.5 (87.3–100.0)	0.59 (0.52–0.66)	27.3 (16.4–40.0)
Noncommunicable disease	1,243	98.6 (98.0–99.2)	0.75 (0.74–0.77)	52.7 (50.2–55.7)
External cause	278	93.9 (91.0–96.4)	0.73 (0.69–0.76)	54.3 (48.6–60.1)
Causes specific to infancy	509	94.3 (92.1–96.3)	0.72 (0.69–0.74)	50.5 (46.2–54.8)
Unspecifiable	256	41.4 (35.9–47.3)	0.24 (0.20–0.28)	9.4 (5.9–12.9)

Panel B. EXP2

Level 1 category	N	Any-code overlap, %	Mean Jaccard	Exact-set agreement, %
Communicable disease	1,781	96.2 (95.3–97.1)	0.62 (0.61–0.64)	31.8 (29.6–33.9)
Direct maternal cause	55	89.1 (80.0–96.4)	0.57 (0.48–0.66)	29.1 (16.4–41.8)
Noncommunicable disease	1,243	94.0 (92.5–95.2)	0.72 (0.70–0.73)	50.7 (47.9–53.3)
External cause	278	82.4 (77.7–86.7)	0.63 (0.59–0.67)	45.7 (40.3–51.4)
Causes specific to infancy	509	80.6 (77.0–84.1)	0.62 (0.59–0.66)	46.4 (41.8–50.9)
Unspecifiable	256	44.9 (39.1–50.8)	0.29 (0.24–0.33)	15.6 (11.7–20.3)

Panel C: EXP3

Level 1 category	N	Any-code overlap, %	Mean Jaccard	Exact-set agreement, %
Communicable disease	1,781	88.5 (87.0–89.9)	0.61 (0.60–0.63)	35.9 (33.6–38.2)
Direct maternal cause	55	89.1 (80.0–96.4)	0.56 (0.47–0.65)	29.1 (18.2–41.8)
Noncommunicable disease	1,243	89.3 (87.6–91.0)	0.75 (0.73–0.77)	61.0 (58.5–63.6)
External cause	278	81.3 (76.6–86.0)	0.70 (0.65–0.74)	58.3 (52.5–64.4)
Causes specific to infancy	509	78.8 (75.2–82.3)	0.62 (0.59–0.66)	46.6 (42.6–50.7)
Unspecifiable	256	40.2 (34.4–46.1)	0.27 (0.23–0.32)	16.4(12.1–21.1)

Panel D: EXP4

Level 1 category	N	Any-code overlap, %	Mean Jaccard	Exact-set agreement, %
Communicable disease	1,781	96.6 (95.7–97.4)	0.63 (0.62–0.64)	32.6 (30.3–34.6)
Direct maternal cause	55	85.5 (76.4–94.5)	0.54 (0.45–0.62)	29.1 (18.2–41.8)
Noncommunicable disease	1,243	93.4 (92.0–94.8)	0.70 (0.68–0.72)	48.2 (45.6–51.0)
External cause	278	82.4 (77.7–86.7)	0.62 (0.58–0.66)	44.2 (38.5–50.4)
Causes specific to infancy	509	83.9 (80.7–87.0)	0.65 (0.62–0.68)	47.3 (43.2–51.9)
Unspecifiable	256	49.2 (43.8–55.5)	0.30 (0.26–0.34)	14.1 (9.8–18.4)

Panel E: EXP5

Level 1 category	N	Any-code overlap, %	Mean Jaccard	Exact-set agreement, %
Communicable disease	1,781	96.3 (95.5–97.1)	0.64 (0.63–0.65)	32.5 (30.5–34.7)
Direct maternal cause	55	94.5 (87.3–100.0)	0.67 (0.58–0.75)	41.8 (29.1–54.5)
Noncommunicable disease	1,243	99.0 (98.5–99.5)	0.79 (0.77–0.80)	58.6 (56.0–61.2)
External cause	278	93.5 (90.3–96.4)	0.79 (0.75–0.82)	65.1 (59.7–70.1)
Causes specific to infancy	509	95.1 (93.1–96.9)	0.75 (0.73–0.78)	57.4 (53.0–61.5)

Unspecifiable	256	29.3 (23.8–35.2)	0.19 (0.15–0.23)	9.8 (6.2–13.3)

Supplementary Table 12 Flexible agreement for contributory causes.

Experiment	Level	EXP contributory set present	Any-code overlap	Mean Jaccard	Exact set agreement
EXP1	1	254	88.3%	0.709	53.7%
EXP1	2	253	65.0%	0.376	19.1%
EXP1	3	253	49.8%	0.268	12.1%
EXP2	1	233	79.4%	0.629	46.7%
EXP2	2	233	59.1%	0.331	16.0%
EXP2	3	233	43.2%	0.229	9.7%
EXP3	1	179	57.2%	0.509	44.7%
EXP3	2	179	40.1%	0.299	21.4%
EXP3	3	179	29.6%	0.207	13.2%
EXP4	1	234	81.3%	0.641	47.1%
EXP4	2	234	63.8%	0.318	10.5%
EXP4	3	234	52.1%	0.238	6.2%
EXP5	1	242	83.3%	0.708	58.4%
EXP5	2	242	59.5%	0.390	23.0%
EXP5	3	242	47.9%	0.294	15.6%

Supplementary Table 13 Jaccard similarity bins at Level 2. This excludes records with unspecified Level 1 underlying code. N=3,882 representing all records that contained at least one clinician CoD for underlying, direct and contributory causes. Percentages may not add up to 100% due to rounding.

Panel	EXP	N	0	>0–0.25	>0.25–0.5	>0.5–0.75	>0.75 -1	1
Overall	1	3,882	13%	20%	49%	3%	-	16%
Overall	2	3,882	27%	15%	39%	2%	-	17%
Overall	3	3,882	25%	7%	44%	2%	-	**22%**
Overall	4	3,882	25%	17%	41%	2%	-	15%
Overall	5	3,882	14%	14%	49%	3%	-	21%
Adult	1	2,468	12%	26%	48%	3%	-	12%
Adult	2	2,468	24%	20%	40%	3%	-	14%
Adult	3	2,468	25%	9%	44%	2%	-	**20%**
Adult	4	2,468	22%	22%	41%	3%	-	11%
Adult	5	2,468	13%	18%	49%	4%	-	17%
Child	1	913	15%	12%	47%	2%	-	25%
Child	2	913	22%	10%	39%	1%	-	29%
Child	3	913	24%	4%	41%	1%	-	**29%**
Child	4	913	21%	10%	43%	1%	-	26%
Child	5	913	15%	9%	47%	2%	-	27%
Neonate/ stillbirth	1	501	17%	5%	58%	2%	-	20%
Neonate/ stillbirth	2	501	52%	2%	32%	0%	-	13%
Neonate/ stillbirth	3	501	26%	2%	51%	2%	-	18%
Neonate/ stillbirth	4	501	50%	2%	36%	1%	-	12%

Neonate/ stillbirth	5	501	17%	1%	51%	1%	-	**30%**

Section F: Exact agreement analysis

Supplementary Table 14 Exact Agreement for underlying cases at all levels and for all experiments.

Panel A: Level 1 underlying causes

Cause	EXP1, n (%)	EXP2, n (%)	EXP3, n (%)	EXP4, n (%)	EXP5, n (%)
00 -Unspecifiable	84 (2.0)	144 (3.5)	87 (2.3)	181 (4.4)	67 (1.6)
01-Communicable disease	1,847 (44.8)	2,035 (49.4)	1,601 (41.9)	2,084 (50.6)	1,737 (42.1)
01-Direct maternal cause	110 (2.7)	83 (2.0)	83 (2.2)	72 (1.7)	126 (3.1)
03-Noncommunicable disease	1,374 (33.3)	1,217 (29.5)	1,399 (36.6)	1,151 (27.9)	1,464 (35.5)
04-External cause	273 (6.6)	275 (6.7)	255 (6.7)	245 (5.9)	305 (7.4)
05-Causes specific to infancy	434 (10.5)	365 (8.9)	396 (10.4)	389 (9.4)	423 (10.3)

Panel B: Top 10 LLM-determined Level 2 underlying causes by experiment

Cause	EXP1, n (%)	EXP2, n (%)	EXP3, n (%)	EXP4, n (%)	EXP5, n (%)
Acute febrile illness	920 (22.3)	1,090(26.5)	859 (22.5)	1,114 (27.0)	832 (20.2)
TB/AIDS	763 (18.5)	791 (19.2)	586 (15.3)	846 (20.5)	741 (18.0)
Cardiovascular disorder	524 (12.7)	461 (11.2)	497 (13.0)	416 (10.1)	572 (13.9)
Neoplasm	238 (5.8)	176 (4.3)	223 (5.8)	199 (4.8)	270 (6.6)
Gastrointestinal disorder	219 (5.3)	189 (4.6)	227 (5.9)	160 (3.9)	218 (5.3)
Diarrhoeal disease without fever	118 (2.9)	143 (3.5)	124 (3.2)	111 (2.7)	124 (3.0)
Birth injury/asphyxia	132 (3.2)	63 (1.5)	140 (3.7)	75 (1.8)	113 (2.7)
Prematurity/low birth weight	146 (3.5)	56 (1.4)	124 (3.2)	52 (1.3)	126 (3.1)
Neonatal sepsis	100 (2.4)	90 (2.2)	84 (2.2)	113 (2.7)	115 (2.8)
Specific information missing on VAQ	34 (0.8)	114 (2.8)	69 (1.8)	146 (3.5)	41 (1.0)

Panel C: Top 10 LLM-determined Level 3 underlying causes by experiment

Cause	EXP1, n (%)	EXP2, n (%)	EXP3, n (%)	EXP4, n (%)	EXP5, n (%)
AIDS	611 (14.8)	648 (15.7)	407 (10.7)	696 (16.9)	609 (14.8)
Malaria	400 (9.7)	217 (5.3)	369 (9.7)	247 (6.0)	436 (10.6)
Pneumonia	237 (5.7)	323 (7.8)	187 (4.9)	382 (9.3)	185 (4.5)
Hypertension	199 (4.8)	164 (4.0)	179 (4.7)	97 (2.4)	263 (6.4)
Cerebrovascular disease	155 (3.8)	161 (3.9)	146 (3.8)	170 (4.1)	109 (2.6)

Acute febrile illness: Unspecifiable	73 (1.8)	290 (7.0)	77 (2.0)	246 (6.0)	16 (0.4)
Pulmonary TB	131 (3.2)	139 (3.4)	160 (4.2)	139 (3.4)	101 (2.5)
Diarrhoeal disease without fever	118 (2.9)	143 (3.5)	124 (3.2)	111 (2.7)	124 (3.0)
Birth injury/asphyxia	132 (3.2)	63 (1.5)	140 (3.7)	75 (1.8)	113 (2.7)
Prematurity/low birth weight	146 (3.5)	56 (1.4)	124 (3.2)	52 (1.3)	

Supplementary Table 15 Breakdown of EXP1 for underlying causes by age group for each VA type.

Panel A: Adult

Cause	Total N	12-14	15-24	25-44	45-64	65+
TB/AIDS	632	14%	19%	45%	36%	4%
Cardiovascular disorder	370	5%	3%	1%	11%	32%
Acute febrile illness	189	9%	12%	7%	5%	9%
Neoplasm	157	5%	4%	3%	8%	9%
Gastro intestinal disorder	126	5%	4%	6%	4%	5%
Transport	58	0%	8%	4%	2%	1%
Endocrine disorders	50	5%	1%	1%	2%	3%
Central nervous system disorder	38	5%	5%	2%	1%	1%
Respiratory disorder	32	0%	2%	1%	2%	1%
Poisoning/exposure to noxious substance	24	0%	2%	1%	1%	0%

Panel B: Child

Cause	Total N	0–1	2–4	5–8	9–11	Out of range
Acute febrile illness	425	46%	52%	40%	36%	17%
TB/AIDS	72	8%	7%	10%	8%	25%
Nutritional disorder	33	3%	5%	5%	3%	0%
Diarrhoeal disease without fever	26	5%	2%	0%	0%	0%
Drowning/submersion	25	0%	3%	8%	5%	8%
Neoplasm	9	0%	2%	1%	8%	0%
Transport	9	0%	1%	4%	5%	0%
Respiratory disorder	7	0%	1%	1%	0%	0%
Gastrointestinal disorder	6	0%	1%	0%	5%	0%
Central nervous system disorder	5	0%	1%	0%	3%	0%

Panel C: Neonate/stillbirths

Cause	Total N	Neonates/Stillborths
Prematurity/low birth weight	122	24%
Birth injury/asphyxia	119	24%
Neonatal sepsis	65	13%
Congenital abnormality	9	2%
Acute febrile illness	8	2%
Unspecifiable	2	0%
Eclampsia	1	0%
Nutritional disorder	1	0%
Sudden infant death syndrome	1	0%

Supplementary Table 16 Experiment 1 against clinician underlying CoDs at Level 2 showing exact matches (true positives) at top clinician categories.

Panel: Adult

Cause	Clinician N	Exact matches	Missed	Sensitivity	Clinician share	Exact-match share	Gap
TB/AIDS	767	632	135	82%	31.3%	25.8%	5.5 pp
Cardiovascular disorder	449	370	79	82%	18.3%	15.1%	3.2 pp
Acute febrile illness	254	189	65	74%	10.4%	7.7%	2.6 pp
Gastrointestinal disorder	196	126	70	64%	8.0%	5.1%	2.9 pp
Neoplasm	183	157	26	86%	7.5%	6.4%	1.1 pp
Endocrine disorders	73	50	23	68%	3.0%	2.0%	0.9 pp
Central nervous system disorder	65	38	27	58%	2.6%	1.5%	1.1 pp
Transport	60	58	2	97%	2.4%	2.4%	0.1 pp
Poisoning/exposure	48	24	24	50%	2.0%	1.0%	1.0 pp
Unspecified non-communicable cause	44	0	44	0%	1.8%	0.0%	1.8 pp

Panel B: Child

Cause	Clinician N	Exact matches	Missed	Sensitivity	Clinician share	Exact-match share	Gap
Acute febrile illness	501	425	76	85%	54.9%	46.6%	8.3 pp
TB/AIDS	142	72	70	51%	15.6%	7.9%	7.7 pp
Nutritional disorder	46	33	13	72%	5.0%	3.6%	1.4 pp
Diarrhoeal disease without fever	36	26	10	72%	3.9%	2.9%	1.1 pp
Drowning/submersion	25	25	0	100%	2.7%	2.7%	0.0 pp
Anaemia	23	5	18	22%	2.5%	0.5%	2.0 pp
Congenital abnormality	17	5	12	29%	1.9%	0.5%	1.3 pp
External cause, unspecified	14	0	14	0%	1.5%	0.0%	1.5 pp
Gastrointestinal disorder	12	6	6	50%	1.3%	0.7%	0.7 pp
Central nervous system disorder	12	5	7	42%	1.3%	0.5%	0.8 pp

Panel C: Neonates/Stillbirth

Cause	Clinician N	Exact matches	Missed	Sensitivity	Clinician share	Exact-match share	Gap
Birth injury/asphyxia	205	119	86	58%	40.9%	23.8%	17.2 pp
Prematurity/low birth weight	145	122	23	84%	28.9%	24.4%	4.6 pp
Neonatal sepsis	96	65	31	68%	19.2%	13.0%	6.2 pp
Neonatal cause, unspecified	20	2	18	10%	4.0%	0.4%	3.6 pp
Acute febrile illness	16	8	8	50%	3.2%	1.6%	1.6 pp
Congenital abnormality	11	9	2	82%	2.2%	1.8%	0.4 pp
Sudden infant death syndrome	2	1	1	50%	0.4%	0.2%	0.2 pp
Nutritional disorder	2	1	1	50%	0.4%	0.2%	0.2 pp

Supplementary Table 17 Showing how records with clinician determinations in the category CD: TB/AIDS: AIDS were classified by EXP1.

EXP1 Cause of death determinations	**Count, %**
CD: Acute febrile illness: Malaria	16 (2%)
CD: Acute febrile illness: Menigitis	9 (1%)
CD: Acute febrile illness: Diarrhoea with fever	6 (1%)
CD: Acute febrile illness: UNSPEC	5 (1%)
CD: Acute febrile illness: Pneumonia	4 (1%)
CD: Acute febrile illness: Other specific unlisted	2 (0%)
CD: Acute febrile illness: Sepsis	1 (0%)
CD: TB/AIDS: AIDS	511 (76%)
CD: TB/AIDS: Pulmonary TB	41 (6%)
CD: TB/AIDS: Extrapulmonary TB	7 (1%)
CD: TB/AIDS: UNSPEC	1 (0%)
CD: Diarrhoeal disease without fever	10 (2%)
CD: Other specific unlisted	8 (1%)
MAT: Puerperal sepsis	1 (0%)
NCD: Cardiovascular disorder: Other specific unlisted	5 (1%)
NCD: Cardiovascular disorder: Ischaemic heart disease	3 (0%)
NCD: Cardiovascular disorder: Cerebro vascular disease	3 (0%)
NCD: Cardiovascular disorder: Congestive heart disease	2 (0%)
NCD: Cardiovascular disorder: Hypertension	1 (0%)
NCD: Respiratory disorder: Asthma	1 (0%)
NCD: Gastro intestinal disorder: Liver cirrhosis	6 (1%)
NCD: Gastro intestinal disorder: Peptic ulcer disease	4 (1%)
NCD: Gastro intestinal disorder: Acute abdomen including obstruction	2 (0%)
NCD: Gastro intestinal disorder: UNSPEC	1 (0%)
NCD: Endocrine disorders: Diabetes	1 (0%)
NCD: Neoplasm: Other specific unlisted	8 (1%)
NCD: Neoplasm: Gastro intestinal tract including abdominal but excluding liver	4 (1%)
NCD: Neoplasm: Oesophagus	3 (0%)
NCD: Neoplasm: Cervix/uterus	2 (0%)
NCD: Neoplasm: Liver	1 (0%)
NCD: Neoplasm: Lung	1 (0%)
NCD: Genito urinary disorders: Kidney disorder	4 (1%)
NCD: Other specific unlisted	1 (0%)

Section G: Chance-Corrected Concordance (CCC) and Cause-specific mortality fraction (CSMF) accuracy

For cause j, first we calculate cause-specific concordance, which is equivalent to sensitivity:

$$C_j = \frac{TP_j}{TP_j + FN_j}$$

Then correct this for the concordance expected from random assignment among N causes:

$$CCC_j = \frac{C_j - \frac{1}{N}}{1 - \frac{1}{N}}$$

where:

- TP_j: clinician cases assigned cause j that the experiment classified as j
- FN_j: clinician cases assigned cause j that the experiment classified as another cause or left blank
- N: number of admissible causes in that cause list

The paper's recommended overall individual- Level measure is the unweighted average across causes. Every cause therefore receives equal importance, regardless of how common it is. This makes average CCC less dependent on the particular cause distribution than accuracy, kappa, or weighted F1. Negative CCC values are possible and mean performance worse than equal-probability random assignment.

The overall CSMF accuracy is calculated using the formula:

$$CSMF\ Accuracy\ = 1 - \frac{\sum_j |CSMF_j^{LLM} - CSMF_j^{Physician}|}{2(1 - min_j CSMF_j^{Physician})}$$

Where for the cause specific CSFM, j is a cause, D_j is the number of deaths assigned to that cause, and N is the total number of deaths.

$$CSMF_j = D_j/N$$

The numerator measures the total absolute difference between the two mortality distributions. CMSF accuracy of 1 means that the LLM reproduces the clinician's cause distribution perfectly, and lower values mean progressively greater differences in the population- Level distribution.

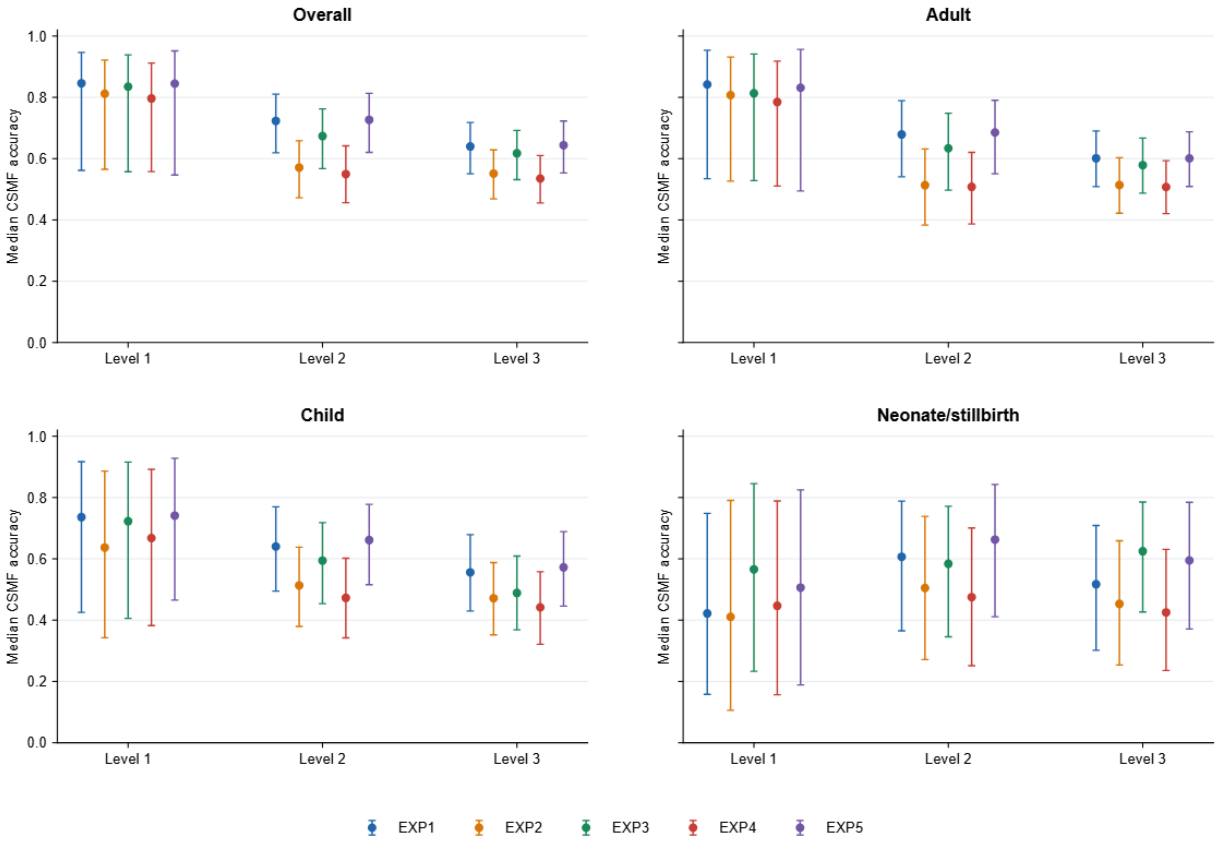
Supplementary Table 18 Robustness of LLM-derived CSMF accuracy to changes in population cause composition.

panel	level	Exp.	Clinician eligible records	N fixed clinician causes	Median CSMF 95 UI
Overall	1	EXP1	4122	6	0.846 (0.562–0.946)
		EXP2	4122	6	0.812 (0.565–0.922)
		EXP3	4122	6	0.835 (0.557–0.939)

		EXP4	4122	6	0.796 (0.558–0.912)
		EXP5	4122	6	0.845 (0.547–0.952)
	2	EXP1	3866	39	0.723 (0.619–0.810)
		EXP2	3866	39	0.571 (0.473–0.658)
		EXP3	3866	39	0.673 (0.568–0.762)
		EXP4	3866	39	0.550 (0.457–0.642)
		EXP5	3866	39	0.727 (0.620–0.813)
	3	EXP1	3866	68	0.640 (0.551–0.718)
		EXP2	3866	68	0.551 (0.469–0.629)
		EXP3	3866	68	0.618 (0.532–0.692)
		EXP4	3866	68	0.535 (0.455–0.610)
		EXP5	3866	68	0.644 (0.553–0.723)
Adult	1	EXP1	2684	5	0.842 (0.535–0.953)
		EXP2	2684	5	0.807 (0.527–0.932)
		EXP3	2684	5	0.813 (0.528–0.941)
		EXP4	2684	5	0.785 (0.510–0.918)
		EXP5	2684	5	0.831 (0.494–0.956)
	2	EXP1	2453	31	0.679 (0.541–0.789)
		EXP2	2453	31	0.513 (0.383–0.631)
		EXP3	2453	31	0.634 (0.497–0.748)
		EXP4	2453	31	0.508 (0.387–0.620)
		EXP5	2453	31	0.685 (0.550–0.790)
	3	EXP1	2453	59	0.601 (0.509–0.691)
		EXP2	2453	59	0.514 (0.422–0.603)
		EXP3	2453	59	0.579 (0.487–0.667)
		EXP4	2453	59	0.508 (0.421–0.593)
		EXP5	2453	59	0.601 (0.509–0.688)
Child	1	EXP1	934	5	0.736 (0.425–0.917)
		EXP2	934	5	0.637 (0.343–0.886)
		EXP3	934	5	0.723 (0.406–0.916)
		EXP4	934	5	0.668 (0.382–0.892)
		EXP5	934	5	0.741 (0.465–0.928)
	2	EXP1	912	28	0.640 (0.495–0.770)
		EXP2	912	28	0.513 (0.379–0.638)
		EXP3	912	28	0.594 (0.454–0.718)
		EXP4	912	28	0.473 (0.342–0.602)
		EXP5	912	28	0.661 (0.516–0.777)
	3	EXP1	912	41	0.556 (0.430–0.679)
		EXP2	912	41	0.471 (0.352–0.588)
		EXP3	912	41	0.489 (0.368–0.609)
		EXP4	912	41	0.442 (0.321–0.558)
		EXP5	912	41	0.572 (0.446–0.689)
Neonate/still birth	1	EXP1	504	6	0.422 (0.158–0.748)
		EXP2	504	6	0.411 (0.106–0.791)
		EXP3	504	6	0.566 (0.233–0.846)
		EXP4	504	6	0.447 (0.157–0.789)

		EXP5	504	6	0.506 (0.188–0.825)
	2	EXP1	501	12	0.607 (0.365–0.788)
		EXP2	501	12	0.505 (0.271–0.739)
		EXP3	501	12	0.584 (0.346–0.772)
		EXP4	501	12	0.475 (0.251–0.701)
		EXP5	501	12	0.663 (0.411–0.843)
	3	EXP1	501	15	0.517 (0.301–0.709)
		EXP2	501	15	0.453 (0.253–0.659)
		EXP3	501	15	0.625 (0.427–0.785)
		EXP4	501	15	0.425 (0.235–0.631)
		EXP5	501	15	0.595 (0.371–0.784)

The table reports median CSMF accuracy and 95% uncertainty intervals across 5,000 alternative cause compositions drawn from a symmetric Dirichlet distribution (α=1) per clinician-supported cause), by experiment, coding Level and VA type. Simulated LLM assignments preserve the observed cause-specific classification behaviour, including blank and out-of-list assignments. The LLMs were not rerun, and results therefore assume that P(LLM-CoD | Clinician CoD) remains stable as the population cause distribution changes. Higher CSMF accuracy indicates closer agreement with clinician-assigned population cause fractions.



Supplementary Figure 1 Robustness of LLM-derived cause-specific mortality fraction accuracy across alternative population cause compositions.

Points show median CSMF accuracy across 5,000 populations generated from a symmetric uninformative Dirichlet distribution (α=1) per clinician-supported cause); error bars show the

2.5th–97.5th percentile uncertainty interval. Results are presented for EXP1–EXP5 at coding Levels 1–3, overall and by VA type. LLMs were not rerun; experimental assignments were sampled from the observed conditional distribution of LLM classifications given the clinician-assigned cause. Higher values indicate better agreement with clinician CSMFs.

Section H: Quality check for a sample of reconstructed narratives

We conducted a sample-based quality assessment of the reconstructed narratives and questionnaire-as-text representations. A sample of 101 records was independently reviewed by two reviewers to assess factual consistency and information preservation. Overall, the reconstructed narratives showed a high degree of factual consistency with the original questionnaire data. No instances were identified in which the LLM introduced unsupported diagnoses or causes of death. The narratives contained no internal logical inconsistencies and preserved important circumstances surrounding death, including injury, pregnancy, delivery and treatment history. Demographic information, such as age and sex, was also accurately represented. However, not all questionnaire variables were explicitly retained in the reconstructed narratives. Coverage of positively recorded responses was high, with each record achieving over 80% complete and accurate coverage. Approximately 65% of the records omitted at least one negative response, although some of these omissions were implicitly covered by general statements indicating the absence of other symptoms. A small number of factual inaccuracies were identified. In three records, all involving children, the LLM incorrectly stated that the child had received a vaccination. One record contained an incorrect symptom duration, while 24 records contained incorrect interpretations of one or two symptoms, while the rest of the symptoms were correctly captured. Some omitted questionnaire variables did not directly describe medical conditions or symptoms. These included questions concerning access to healthcare, such as treatment costs and travel distances, which had been introduced in later versions of the questionnaire.

Supplementary Table 19 Sample quality check for narratives reconstruction.

Question	Yes, N	Yes, %	Positives correctly captured/ Total	%	Negatives correctly captured/ Total	%
1. Does the edited narrative contain any additional factual information that is not present in the unedited narrative?	—	—	0 / 0	—	0 / 0	—
2. Does the edited narrative introduce any diagnosis or suspected cause of death that is not explicitly stated in the unedited narrative?	—	—	0 / 0	—	0 / 0	—
3. Does the edited narrative incorrectly interpret any information in the unedited narrative?	24	24%	420 / 535	79%	691 / 996	69%
4. Does the edited narrative contain contradictory statements?	—	—	0 / 0	—	0 / 0	—

5. Are any symptoms or conditions recorded as positive (Yes) missing from the edited narrative?	53	53%	1,101 / 1,363	81%	1,607 / 2,691	60%
6. Are any symptoms or conditions recorded as negative (No) missing from the edited narrative?	66	65%	1,302 / 1,586	82%	2,199 / 3,526	62%
7. Does the edited narrative mention a general statement such as “no other significant symptoms or conditions were reported”?	16	24%	406 / 499	81%	503 / 1,224	41%
8. Does the edited narrative omit any important circumstances surrounding the death, such as injury, pregnancy, delivery or treatment history?	—	—	0 / 0	—	0 / 0	—
9. Does the edited narrative incorrectly report the duration, frequency or severity of any symptom?	1	1%	22 / 31	71%	35 / 71	49%
10. Does the edited narrative change an uncertain, unknown or missing response into a definite positive or negative statement?	3	3%	21 / 26	81%	51 / 53	96%
11. Does the edited narrative incorrectly report demographic information, such as age or sex?	—	—	0 / 0	—	0 / 0	—
All reviews	**101**	**100%**	**1,649 /1,991**	**83%**	**2,893 /4,234**	**68%**

Section I: Examples of misclassified records

Example of narratives for neonate/stillbirth VAs that were coded as obstructed labour by the LLM and as birth injury/asphyxia by the clinician:

> Example1: *The mother reports that the baby was delivered with difficulties. It was a breech presentation. She says when the legs were delivered the head was stuck in pelvis for a very long time and later get delivered, and the baby had very severe difficulties in breathing.*
>
> Example 2: *The deceased was a full-term baby at the time of birth, but showed very poor signs of life e.g. the baby could not cry and started breathing after several resuscitation measures. The baby died three hours later suggestively by the guardians due to long hour of labour.*

In the first example, the narrative contains clear evidence of obstructed labour, hence either of the two causes may be selected either as underlying or direct. In the second example, the narrative does not mention obstructed labour. In this case, the LLM provided helpful determinations, but would require clinicians to apply the current guidance and practice to balance alternative interpretations.